

Faculté des Sciences de Rabat

Master IPS

Master's/PFE Report

LLM-as-a-Judge Reliability Lab: Measuring and Mitigating Biases in Automatic Evaluation of Generated Responses

A reproducible empirical study of agreement with human preference reference labels, repeated-evaluation consistency, answer-order sensitivity, controlled presentation effects, source-family association, and mitigation trade-offs.

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Abstract

This report presents the *LLM-as-a-Judge Reliability Lab*, a reproducible study of whether large language models can act as dependable pairwise evaluators of generated responses. The central premise is deliberately modest: an evaluator verdict is an observation produced by a model, a prompt, a presentation order, and a protocol. It is not a ground-truth statement. The study therefore measures several distinct reliability properties instead of asking whether one judge is “best.”

The final analysis uses a source-text-corrected controlled dataset with 1,568 records. It represents 80 distinct MT-Bench questions (upstream question IDs 81–160) as 160 turn-level prompts. Human preference reference labels are retained as the comparison target. The source-corrected evidence uses complete-case estimators, fixed seeds, 10,000 bootstrap iterations, no imputation, and provenance-preserving source precedence. Fourteen terminal Claude observations remain unavailable and are transparently excluded where a complete case is required.

Agreement with human preference reference labels was 59.20% (457/772; 95% CI 55.70–62.69%; $\kappa = 0.3230$). Repeated evaluations under fixed conditions were highly consistent (96.53%, 7,119/7,375). In contrast, answer-order presentation mattered: 17.49% of decisive paired comparisons changed their mapped outcome after an order swap (96/549). The narrowly defined redundant-text and presentation-format treatments produced no stable variant wins in their eligible pairs (0/682 and 0/716 respectively); these results are not general claims about verbosity or formatting. A counterbalanced source-family experiment estimated a 50.94% stable same-family preference rate (81/159; 95% CI 42.77–58.49%) with strong judge-level heterogeneity.

Two mitigation families were evaluated as complementary, not head-to-head, strategies. The primary DUAL_SWAP within-judge presentation-consistency filter had a +0.67 percentage-point agreement change (95% CI -0.17 to +1.68) on its retained matched population, while reducing coverage by 21.03 points. The secondary strict four-judge consensus protocol

improved agreement by +5.02 points (95% CI +4.16 to +5.89) against its own equal-weight individual-judge comparator, retaining 69.83% of planned pairs. The report documents the methods, findings, limitations, software architecture, and an evidence-led reproducibility workflow.

Keywords: LLM-as-a-Judge; reliability; human preference reference labels; position sensitivity; reproducibility; consensus; controlled evaluation.

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Chapter 1

Introduction

Generative systems increasingly produce answers, summaries, code explanations, and conversational assistance at a scale that makes manual evaluation expensive. Pairwise evaluation by a language model is attractive because a judge can read a task, compare two candidate answers, and return a compact preference. This apparent simplicity is also the central methodological risk. A verdict may change because the evaluator sees a different order, because a response has superficial presentation features, because repeated sampling is not identical, or because the protocol filters observations in ways that change coverage. A useful evaluator therefore needs to be studied as an empirical measurement instrument.

The LLM-as-a-Judge Reliability Lab treats each verdict as a recorded experimental observation rather than as an unquestionable assessment. The project asks how closely judges agree with human preference reference labels, whether they repeat their decisions under fixed conditions, whether presentation order affects their decisions, whether narrowly controlled text variants receive a systematic advantage, whether source-family identity is associated with preference in a counterbalanced design, and what mitigation strategies gain or lose. This framing avoids a false binary between “works” and “does not work.” A judge can be stable in one setting, sensitive in another, and operationally useful only with clearly stated limits.

1.1 Problem statement

Automatic evaluation becomes scientifically fragile when an evaluator’s output is reported without the provenance needed to interpret it. In particular, a pairwise verdict depends on the exact prompt, the exact two answer texts, their original source identities, the

evaluator model, sampling and parser settings, and the ordered protocol. If any of these identities is reconstructed from display labels or database order, the displayed evidence can become incorrect even when the surrounding interface looks plausible. The project therefore separates answer-source provenance from judge identity, records decision attempts and terminal states, and pins final metrics to authoritative analysis runs.

The final study also responds to a data-lineage issue discovered during the project. The controlled population was rebuilt against exact upstream source text. The result is an additive source-corrected evidence lineage: 1,568 records, 1,568/1,568 source hashes matched, and zero unresolved records. The project does not overwrite historical artifacts. Instead, it distinguishes historical evidence from the source-corrected final analysis and makes the final selection explicit. This is important because a reliable numerical result should also be traceable to the text that a judge actually evaluated.

1.2 Objectives

The practical objective is a local research application that can store controlled executions, expose final results safely, and distinguish final evidence from exploratory telemetry and manual sandbox trials. The scientific objective is to quantify several dimensions of evaluator reliability under fixed, transparent protocols. The reproducibility objective is to make the chain from canonical data to releases, analysis runs, API output, and interface visible and verifiable without performing new provider calls.

This combination creates a different kind of deliverable from a benchmark leaderboard. The software is useful for operating the study, but the software alone is not the claim. The claims are the estimands, denominators, confidence intervals, exclusions, and evidence identities reported in this document. Conversely, a table of percentages alone is not enough: the analysis must remain reconstructible from stored records and frozen artifacts.

1.3 Research questions

The seven research questions are deliberately separated because they measure different phenomena. RQ1 measures agreement with human preference reference labels. RQ2 measures repeated-evaluation consistency under fixed conditions. RQ3 measures position sensitivity by pairing both answer orders and mapping the results back to original answer identity. RQ4 measures a controlled redundant-length effect, not generic verbosity. RQ5

measures a controlled presentation-format effect, not generic formatting quality. RQ6 measures a counterbalanced source-family preference association. RQ7 evaluates mitigation trade-offs using two complementary strategies.

Table 1.1: Canonical research questions and principal estimands.

RQ	Focus	Principal final estimand
RQ1	Human alignment	Exact three-class agreement with human preference reference labels; Cohen’s κ .
RQ2	Stochastic consistency	Strict consistency across repeated fixed-condition evaluations.
RQ3	Position sensitivity	Paired decisive mapped flip rate under AB/BA answer-order swaps.
RQ4	Redundant length	Stable win rate of a narrowly defined redundant-text variant.
RQ5	Presentation format	Stable win rate of a controlled list-prefix/presentation variant.
RQ6	Source family	Stable same-family preference rate in a counter-balanced lineage.
RQ7	Mitigation	Agreement and coverage trade-offs under DUAL_SWAP and strict multi-judge consensus.

1.4 Contributions

The work contributes an evidence-oriented controlled evaluation workflow, an explicit provenance model, a source-text correction and recovery lineage, and a complete final analysis that reports both numerator/denominator detail and uncertainty. It also provides a cautious interpretation framework. Agreement with a human preference reference label is not ground truth. A measured order effect is not proof of a causal internal bias mechanism. A null result for a narrow synthetic perturbation is not proof that length or format never matter. A mitigation is not universally better simply because it improves one estimator while reducing coverage.

1.5 Report roadmap

Chapter 2 reviews the relevant literature and derives the research gap. Chapter 3 describes the data, architecture, provenance, and correction lineage. Chapter 4 specifies the designs and estimators. Chapter 5 reports the final source-corrected results. Chapter 6 interprets findings and limitations. Chapter 7 documents the software and reproducibility pathway. The appendices provide an evidence protocol, a compact numerical ledger, and practical verification guidance.

Chapter 2

State of the Art and Research Gap

2.1 LLM-based evaluation as a measurement problem

LLM-based evaluation has become an important practical response to the cost of expert annotation. A language model can apply a rubric, compare responses, and provide structured output. The literature, however, does not present this as a solved replacement for human evaluation. It presents a family of methods whose quality depends on task, prompt, model, reference material, and protocol. The appropriate question is therefore not whether an LLM can judge in the abstract, but which reliability properties are observed under a reproducible design.

The MT-Bench and Chatbot Arena study by Zheng et al. [5] made this issue concrete by using strong models as evaluators while examining agreement, position effects, verbosity effects, and self-enhancement concerns. It also established MT-Bench as a useful multi-turn setting for comparative evaluation. The present project adopts the need for a controlled and traceable benchmark context, while focusing on a smaller, locally reproducible set of stored pairwise decisions and explicitly separating 80 distinct questions from their 160 turn-level prompt records.

2.2 Alignment with human judgements

Several approaches seek stronger correspondence with human assessment. G-EVAL uses a structured prompting procedure with chain-of-thought and form-filling ideas to evaluate natural-language generation, reporting improved correlation with human judgements in

its target settings [2]. This direction motivates explicit rubric and output-parsing design. Yet correlation or agreement alone does not establish that an evaluator is robust to order, repetition, or presentation changes. A judge can be aligned on average but still produce unstable or protocol-sensitive individual decisions.

For that reason, the present project treats human labels as *human preference reference labels*. They are the defined comparison target for RQ1, not an assertion of objective correctness. The three-class outcome space includes answer 1, answer 2, and tie. This is an important distinction: collapsing ties or treating all disagreement as one homogeneous type can make a simple percentage look more decisive than the evaluation setting warrants.

2.3 Position and presentation sensitivity

Order sensitivity is among the most visible risks in pairwise evaluation. Shi et al. [3] systematically study position bias across judges, tasks, and candidate models, distinguishing repetition stability, position consistency, and preference fairness. Wang et al. [4] likewise report fairness concerns for large-language-model evaluators and study a calibration-oriented response. These studies motivate a design that does not infer order sensitivity from a single displayed order. Instead, the project evaluates AB and BA passes and maps both decisions to the original answer identities before calculating paired outcome disagreement or decisive flips.

Presentation sensitivity can be confused with content sensitivity. The literature on length control provides a related example: Dubois et al. [1] argue that automatic evaluators can be confounded by answer length and propose a controlled adjustment procedure. Their work motivates experimental control, but it does not license a universal inference from one perturbation. RQ4 therefore asks only whether a specifically frozen redundant-text variant receives a stable win in the present population. RQ5 similarly asks only whether a frozen list-prefix/presentation treatment receives a stable win. Neither estimand claims to settle all effects of answer length or formatting.

2.4 Mitigation and aggregation

Mitigation approaches vary in what they change. Some modify prompts or scoring rules; others filter observations; others aggregate multiple judges. The present project distinguishes two families because their outputs are not directly comparable as if they had the

same units and comparator. DUAL_SWAP is a within-judge presentation-consistency filter. It retains cases satisfying a fixed mapped-order criterion and therefore changes coverage. Strict multi-judge consensus aggregates independent judge votes and compares consensus against an equal-weight individual-judge baseline on the same retained pairs. It is a secondary RQ7 strategy, not a replacement for DUAL_SWAP.

This distinction matters scientifically. A multi-judge agreement change measured on a strict four-judge retained population cannot be subtracted from a DUAL_SWAP agreement change measured on a different retained matched population to claim superiority. Both are valuable results, but they answer different operational questions. The report therefore preserves the comparison boundary in its tables and interface description.

2.5 Research gap

The literature identifies the central risks, but a research project also needs to show how those risks are managed end to end. The gap addressed here is an integrated, reproducible workflow that joins: (i) exact source-text provenance, (ii) controlled RQ-specific designs, (iii) stored operational terminal states, (iv) analysis-run-pinned API values that fail closed when a required artifact is absent or ambiguous, and (v) cautious claims tied to a named frozen estimator. The project also makes a practical contribution by separating exploratory historical telemetry, final controlled evidence, and live manual trials so that a convenient interface cannot silently contaminate a final conclusion.

2.6 Conceptual framework

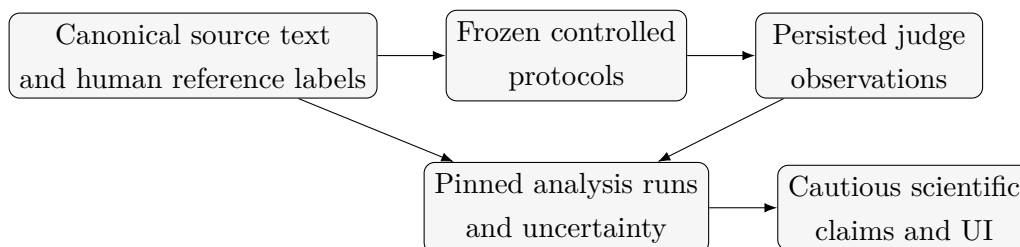


Figure 2.1: The project treats reliability claims as a chain from exact source text through a frozen protocol and persisted observations to a pinned analysis.

The framework is intentionally conservative. Every link can fail independently: source text may be mismatched, a provider response may be invalid, an observation may be terminally unavailable, a repeated pass may be incomplete, or an analysis identity may be missing. A reliable system records these conditions and fails closed rather than silently manufacturing a result. This design guides the rest of the report.

Chapter 3

Study Design, Data, and Provenance

3.1 Study population

The final controlled population is a source-text-corrected dataset version named `controlled-source-text`. It contains 1,568 records derived from 80 distinct MT-Bench questions, retaining upstream question IDs 81 through 160. MT-Bench is multi-turn; each question is represented by two turn-level prompt records. Thus, “160 prompts” in this report always means 80 questions multiplied by two turns, not 160 distinct questions. The canonical dataset hash is `f4dd4b2ec9e44d61b7e963b34eba6fe796f61fbe930100f9c6fff82da6ab1209`.

The human comparison dataset was reconciled against upstream records: 3,355 of 3,355 raw human rows matched. In the canonical controlled source data, 1,070 records were exact source matches and 498 required source correction; unresolved records equal zero. These counts are not a performance result. They document the population and source lineage on which all later estimators depend.

Table 3.1: Canonical data lineage summary.

Item	Count / status
Distinct MT-Bench questions	80 (IDs 81–160)
Turn-level prompt records	160 (80 turn 1; 80 turn 2)
Canonical source-corrected records	1,568
Exact source records	1,070
Source-correction-required records	498
Unresolved source records	0
Human rows reconciled with upstream	3,355 / 3,355

3.2 Provenance model

Each controlled observation is associated with exact prompt and answer identities, source-model identities, a judge identity, a protocol condition, and a durable attempt/slot state. The system must not obtain answer text merely by using a prompt identifier and whichever two answers appear first in a database query. Such an approach can relabel one source model’s text as another source model’s answer. The project therefore treats answer provenance and judge selection as separate fields. This separation is also used by the qualitative explorer: the user-selected top-level model filters the judge perspective, whereas answer A and answer B retain their exact source identities.

The evidence pipeline applies a stated source precedence: source-correct reusable observations first, then corrected original observations, then recovery replacements, and finally terminal missing states. The analysis does not impute unavailable outcomes, substitute another model, or relax the decision parser merely to increase coverage. This policy turns missingness into a visible limitation rather than an invisible data-cleaning choice.

3.3 Operational accounting

The complete source-corrected controlled ledger contains 17,560 logical controlled passes. Of these, 13,103 are source-correct reusable, 4,373 corrected original slots, and 84 recovery replacements. Nine logical controlled passes are terminally missing; the total across all

lineages includes fourteen terminal missing Claude observations. The distinction between physical, valid, and analytical counts is maintained per research question. A physical pair can exist even when it is not eligible for a decisive or complete-case estimator.

3.4 Software architecture

The application uses a React/Vite frontend and a FastAPI backend backed by a relational database. The backend is organized around core configuration and models, data construction, evaluation protocols and persistence, analysis, specialized experiments, multi-judge execution/analysis, release tooling, and historical audit lineage. The final controlled-results route obtains values from authoritative analysis runs rather than from display constants. If the unique required result or provenance identity is missing, duplicated, or malformed, the route fails closed instead of returning an invented numerical fallback.

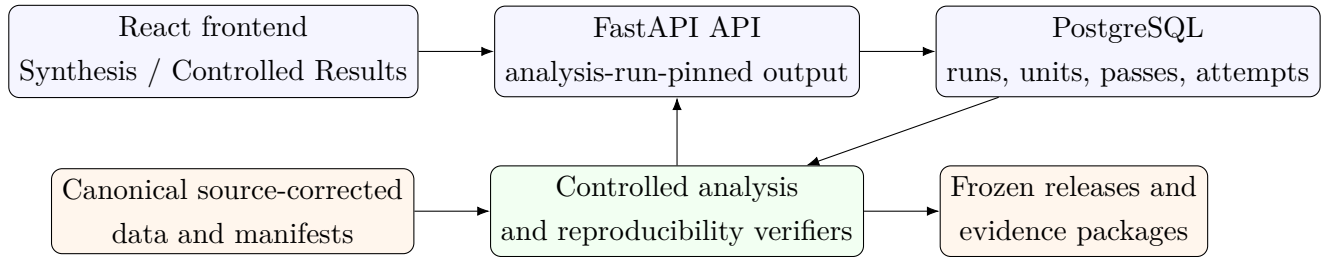


Figure 3.1: High-level architecture. Final values flow from identifiable analysis outputs; exploratory and live sandbox data are not a fallback source for controlled results.

3.5 Evidence separation

The application exposes six user-facing areas. The default Leaderboard, Bias Diagnostics, and Qualitative Explorer are historical exploratory telemetry views. The Synthesis and Controlled Experiments views expose final controlled evidence from the final API. Live Evaluation is a manual single-trial sandbox. It can be disabled by default through a server-side environment gate and is not part of the frozen RQ1–RQ7 evidence. A live result therefore cannot be described as a population-level bias finding, a proof of debiasing, or a new final experimental result.

3.6 Final evidence identity

The source-corrected complete-case full-population analysis is identified by SHA-256 `ace75cfb15412c2070b848f048ad419ae70d46162c7b66d52ee4302432031d97`. Its analysis version is `source-corrected-complete-case-full-population-analysis-v2`. The final RQ results are pinned to eight authoritative analysis-run identifiers, including primary and secondary RQ7 runs. This report cites these identities in the evidence appendix rather than relying on an informal statement that a number is “final.”

Chapter 4

Experimental Methods

4.1 Common conventions

All controlled questions are assessed using a three-class decision space: original answer 1, original answer 2, or tie. A parser may also encounter an invalid or unavailable response, but invalid and terminal states are not silently treated as ties. For an estimator requiring a complete or decisive case, the denominator is reported explicitly. The project uses deterministic analysis seeds and 10,000 bootstrap iterations for confidence intervals. Unless noted otherwise, uncertainty intervals are non-parametric bootstrap 95% intervals from the frozen analysis.

The study does not run a new provider call during analysis. It reads persisted observations with their state and provenance. This distinction is useful because an analysis result remains reviewable even if a provider is unavailable later. It also prevents a rerun with a changed model or routing policy from silently replacing the original evidence.

4.2 RQ1: agreement with human preference reference labels

RQ1 compares a judge outcome with the available human preference reference label on the same pair. Exact agreement is the proportion of matching three-class outcomes. Cohen’s κ is reported as a chance-corrected companion statistic. The denominator excludes observations that cannot be validly classified under the frozen policy. The result should be interpreted as agreement with a reference labeling process, not as accuracy against ground

truth.

If J_i is the valid judge outcome and H_i the human reference label for observation i , exact agreement is

$$\hat{A} = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(J_i = H_i).$$

The final RQ1 analytical population is 772 observations. It is smaller than the planned 800 because the analysis preserves validity and complete-case requirements rather than filling unavailable outcomes. Category and judge stratifications are descriptive extensions, not alternative primary estimands.

4.3 RQ2: fixed-condition repetition consistency

RQ2 repeats the same evaluation under the same frozen condition. It does not estimate the effect of arbitrary sampling temperatures or provider changes. The strict complete estimator asks whether all required repetitions in a complete cell agree exactly. The final result uses 1,475 strict valid cells with 7,375 valid repeated decisions. A conditional sensitivity denominator is separately retained so that readers can see how requiring complete cells changes the population.

Repeated consistency is operationally useful: it measures whether a protocol returns the same outcome when asked again under the same conditions. It should not be interpreted as an intrinsic property of a model across all prompts, all deployment settings, or all future versions. A high repetition score can coexist with substantial position sensitivity, which is exactly why RQ2 and RQ3 are not merged into a single reliability score.

4.4 RQ3: answer-order sensitivity

For each candidate pair, the study obtains an AB pass and a BA pass. The raw slot labels are converted back to the identity of the original answer. A mapped decisive flip occurs if both passes are decisive and their mapped winners differ. This mapping is essential: comparing raw first/second slot labels would confound a deliberate order swap with a preference for an original answer.

Let M_{AB} and M_{BA} be the mapped original-answer outcomes. The decisive flip estimator is

$$\hat{F} = \frac{\sum_i \mathbb{I}(M_{AB,i} \neq M_{BA,i}) \mathbb{I}(M_{AB,i}, M_{BA,i} \in \{A, B\})}{\sum_i \mathbb{I}(M_{AB,i}, M_{BA,i} \in \{A, B\})}.$$

The analysis also retains all-paired outcome disagreement as a descriptive quantity, because ties and invalid outcomes can make the broader paired population larger than the decisive subset. The primary RQ3 result is the decisive mapped flip rate.

4.5 RQ4: controlled redundant-length effect

RQ4 uses a frozen redundant-text treatment. The treatment is intentionally narrow: it changes the defined answer variant in a controlled way while keeping the underlying pair design fixed. A stable treatment win requires the relevant matched outcomes to meet the frozen validity and stability criterion. The numerator counts only cases where the treatment variant wins; the denominator counts eligible controlled pairs.

This design answers a precise question. It does not establish that all longer answers are preferred, that length never matters, or that a judge is generally free from verbosity-related behaviour. The value of a null finding here is methodological: it bounds the claim to the implemented treatment and population.

4.6 RQ5: controlled presentation-format effect

RQ5 parallels RQ4 but uses a frozen presentation/list-prefix treatment. The same discipline applies. The outcome is a stable treatment-variant win rate on eligible controlled pairs, not a general measure of style preference. Reporting the construction, eligibility rule, and denominator is especially important for a result at or near zero because a zero numerator is meaningful only relative to the number and quality of opportunities for it to occur.

4.7 RQ6: counterbalanced source-family association

RQ6 evaluates whether the answer produced by the same source family as the judge is selected in a counterbalanced experiment. The counterbalanced lineage includes 480 planned units and uses stored exact answer/source identities. Its analytical stable decisive

denominator is 159, yielding coverage of 33.13%. The estimator is the proportion of stable decisive outcomes selecting the same-family answer. The counterbalancing design improves interpretability relative to an unbalanced display, but it does not prove a causal self-bias mechanism. Judge-level heterogeneity is reported because an aggregate near 50% can mask very different patterns across judges.

4.8 RQ7: complementary mitigation strategies

RQ7 contains two separately defined mitigation families. The primary DUAL_SWAP protocol is a within-judge presentation-consistency filter. It compares baseline and DUAL_SWAP agreement on its own linked retained population and separately reports coverage relative to complete linked triplets. Because the protocol filters cases, a higher agreement alone would not be enough; the coverage change is part of the result.

The secondary Multi-Judge Consensus protocol aggregates strict four-judge votes. It retains pairs satisfying its own validity condition and compares the consensus outcome with an equal-weight individual-judge baseline on exactly the same retained pairs. Its planned population, retained population, comparator, and confidence interval are therefore distinct from DUAL_SWAP. No direct head-to-head delta is defined between the two methods.

Table 4.1: Final analysis populations by research question.

RQ	Estimator population	Planned	Analytical	Interpretation boundary
RQ1	valid human-label comparisons	800	772	reference-label agreement
RQ2	strict complete repetition cells	1,600	1,475	fixed-condition repetition
RQ3	decisive mapped AB/BA pairs	800	549	mapped order flips
RQ4	eligible stable redundant-text pairs	800	682	one frozen treatment
RQ5	eligible stable format pairs	800	716	one frozen treatment
RQ6	stable decisive counterbalanced units	480	159	association; heterogeneity
RQ7 primary	retained linked DUAL_SWAP units	800	597	within-judge filter
RQ7 secondary	strict consensus retained pairs	1,611	1,125	cross-judge aggregation

4.9 Statistical interpretation

Point estimates are reported with their denominators and bootstrap intervals where the final artifact provides them. The purpose of an interval is not merely to decorate a percentage. It communicates how much the observed sample constrains the estimator under the frozen resampling procedure. A confidence interval crossing zero for a matched difference is a reason to avoid a definitive positive/negative claim, even if the point estimate is positive. The project deliberately does not collapse all RQs into a single aggregate score. Agreement, repeated consistency, order sensitivity, controlled treatment response, source-family association, coverage, and consensus behaviour have different units and failure modes. Combining them would hide the trade-offs that a responsible evaluator deployment must expose.

Chapter 5

Final Source-Corrected Results

5.1 Evidence basis

All values in this chapter come from the source-corrected complete-case full-population analysis artifact, version `source-corrected-complete-case-full-population-analysis-v2`, SHA-256 `ace75cfb15412c2070b848f048ad419ae70d46162c7b66d52ee4302432031d97`. The evidence selection is explicit because earlier historical lineages are retained for auditability but are comparison-only. No new provider requests were performed to calculate the values reported here.

5.2 RQ1: agreement with human preference reference labels

Across 772 analytical observations, judges exactly matched the human preference reference label in 457 cases, an agreement rate of 59.20% (95% CI 55.70–62.69%). Cohen’s κ was 0.3230 (95% CI 0.2698–0.3758). This is a moderate association in the practical sense that agreement is visibly above a naive random baseline but far from universal. The result must not be rephrased as ground-truth accuracy: human preference labels are the stated reference and can contain task- and annotator-dependent judgements.

The descriptive per-judge rates show heterogeneity: Claude 45.45% (85/187), DeepSeek 67.55% (127/188), GPT 57.00% (114/200), and Llama 66.50% (131/197). These numbers are not a global model ranking. They arise within this controlled dataset, protocol, validity policy, and human-label population.

5.3 RQ2: repeated-evaluation consistency

The strict complete repeated-evaluation result is 96.53% (7,119/7,375 valid repeated decisions) across 1,475 strict valid cells. A conditional sensitivity analysis reports 96.31% over 1,585 cells. Physical completeness was 1,599/1,600 cells. Thus the protocol was highly repeatable under its fixed condition, with transparent missingness rather than forced completion. This result does not contradict RQ3: a judge may repeat a decision reliably when shown the same prompt and still react differently when the answer order changes.

5.4 RQ3: position sensitivity

Among 799 physical AB/BA pairs, 759 were valid paired outcomes and 549 were decisive after mapping decisions back to original answer identities. There were 96 decisive mapped flips, giving a primary flip rate of 17.49%. The broader all-paired outcome disagreement rate was $193/759 = 25.43\%$. The gap between these values reflects the distinction between a decisive winner reversal and any mapped outcome difference, including ties.

Per-judge decisive flip rates were Claude 53.70% (58/108), DeepSeek 10.98% (19/173), GPT 2.70% (3/111), and Llama 10.19% (16/157). This is an empirical position-sensitivity observation within the experiment; it does not identify a causal internal mechanism or license a general statement about future model versions.

5.5 RQ4 and RQ5: narrow controlled treatments

RQ4 found zero stable redundant-text variant wins among 682 eligible pairs (0.00%). RQ5 found zero stable presentation/list-prefix variant wins among 716 eligible pairs (0.00%). Both results are clear for the specified frozen treatments and eligible denominators. They should not be turned into broad claims that longer answers, verbose answers, or all formatting changes do not influence language-model judges. The point of these experiments is to make the treatment sufficiently narrow that the inference is honest.

5.6 RQ6: matched source-family preference

The counterbalanced source-family result is $81/159 = 50.94\%$ stable same-family preference (95% CI 42.77–58.49%), with 33.13% coverage from 159 of 480 planned units. The aggregate interval includes 50%. Judge-specific descriptive rates vary strongly: Claude $25/29 = 86.21\%$, GPT $50/66 = 75.76\%$, and Llama $6/64 = 9.38\%$. The evidence therefore does not support a uniform overall same-family preference claim. It supports an observed association in the counterbalanced protocol with strong judge-level heterogeneity and limited stable-decisive coverage.

5.7 RQ7: mitigation trade-offs

For primary DUAL_SWAP, 799 of 800 linked triplets were physically complete and 597 were retained for the matched agreement comparison. Baseline agreement was $405/597 = 67.84\%$; DUAL_SWAP agreement was $409/597 = 68.51\%$. The matched change was +0.67 percentage points with a 95% CI of -0.17 to +1.68 points. Baseline coverage was $774/799 = 96.87\%$, while DUAL_SWAP coverage was $606/799 = 75.84\%$, a -21.03 point change. The appropriate reading is a small, uncertain agreement change paired with a material coverage trade-off.

For secondary strict Multi-Judge Consensus, 1,125 of 1,611 planned pairs were retained (69.83% coverage). Consensus agreement with human preference reference labels was $808/1,125 = 71.82\%$, compared with 66.80% for the equal-weight individual-judge comparator on the same retained pairs. The matched difference was +5.02 points (95% CI +4.16 to +5.89). This supports the limited claim that, under this frozen strict four-judge consensus protocol and its own comparator, aggregation increased reference-label agreement on retained pairs. It does not establish superiority over DUAL_SWAP because the units, comparator, and retained population differ.

Table 5.1: Final primary results. Percentages are reported with their frozen analytical denominators.

RQ	Final result	N	Cautious interpretation
RQ1	59.20% agreement; $\kappa = .3230$	772	agreement with human preference reference labels
RQ2	96.53% strict consistency	1,475 cells	fixed-condition repeatability
RQ3	17.49% decisive mapped flips	549	answer order can change a decisive result
RQ4	0.00% redundant-variant wins	682	no stable win in this frozen treatment
RQ5	0.00% format-variant wins	716	no stable win in this frozen treatment
RQ6	50.94% same-family preference	159	no uniform aggregate preference; heterogeneous
RQ7 DUAL_SWAP	+0.67 pp agreement; -21.03 pp coverage	597	small uncertain gain; lower coverage
RQ7 Multi-Judge	+5.02 pp agreement; 69.83% coverage	1,125	secondary within-own-comparator effect

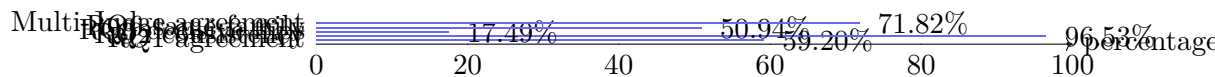


Figure 5.1: Selected final percentages. Metrics measure different estimands and are not a common score or ranking.

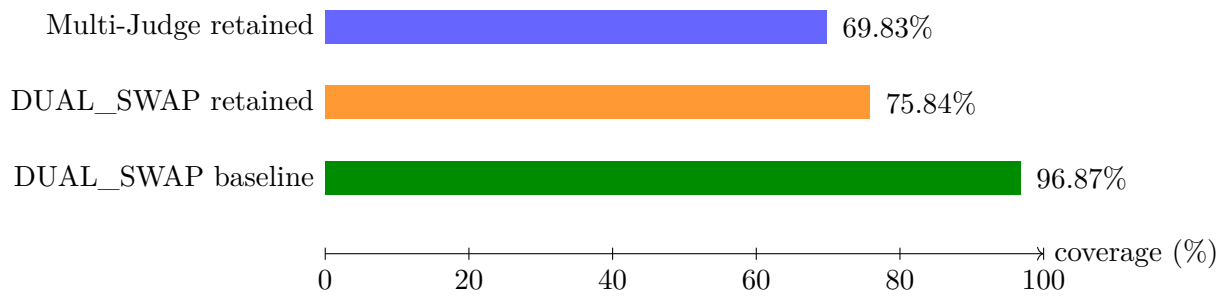


Figure 5.2: Coverage is part of the RQ7 mitigation result. The methods use different retained populations and comparators.

Chapter 6

Discussion

6.1 What the results jointly show

The results describe a judge ecosystem with useful but bounded reliability. The RQ1 agreement level shows that automated judgments capture a meaningful portion of the human preference reference pattern, yet leave a substantial disagreement rate. RQ2 shows that the same frozen prompt/protocol is usually repeated consistently. RQ3 demonstrates why this is not enough: changing only answer order can change a decisive mapped outcome for a non-trivial share of eligible pairs. This combination is precisely why a single global “reliability” label would be misleading.

The RQ4 and RQ5 null results are also informative when interpreted with the right scope. The specified redundant-text and presentation/list-prefix treatments did not produce stable wins in their eligible controlled pairs. That finding limits a claim about these treatments; it does not settle a broader debate about verbosity, style, salience, or prompt-dependent formatting. A scientifically useful null result is not an invitation to say “no bias exists.” It is a well-bounded observation under a frozen intervention.

RQ6 adds another dimension. The aggregate same-family preference rate is close to 50%, and its interval spans 50%, so a uniform overall preference claim is not supported. At the same time, judge-level patterns differ sharply. This is a caution against interpreting aggregate evidence as an explanation of every component judge. It also illustrates why counterbalancing, exact answer identity, and reporting coverage are essential in source-family questions.

6.2 Interpreting mitigation

DUAL_SWAP and Multi-Judge Consensus address different practical concerns. DUAL_SWAP asks whether a within-judge decision remains compatible with a presentation-consistency requirement. It may remove decisions whose evidence is less stable under answer swapping; in doing so it visibly reduces coverage. In the final analysis its matched agreement change is small and its confidence interval crosses zero. This does not make the method useless. It means the particular agreement-gain estimand is uncertain in this population and should always be read with the 21.03 point coverage reduction.

Multi-Judge Consensus aggregates four independent judge votes under a strict frozen protocol. On its own retained pairs, it improves agreement with human preference reference labels by 5.02 points compared with an equal-weight individual-judge baseline. The confidence interval is positive under the frozen analysis. This is evidence for a secondary mitigation result, not evidence that an ensemble is universally better. The protocol excludes pairs without the required valid votes, retains 69.83% of its planned population, and evaluates a cross-judge aggregation family rather than a within-judge filter.

The two results should not be placed on a one-dimensional race track. Their comparators differ. Their retained units differ. Their operational costs and failure modes differ. A future applied deployment may choose a DUAL_SWAP-like filter when the priority is presentation robustness within a judge, or a multi-judge procedure when the priority is aggregation across independent models and the coverage/cost profile is acceptable. That is a decision framework, not a claim of universal superiority.

6.3 Threats to validity

6.3.1 Construct validity

Human preference reference labels are a defined reference target, not ground truth. They may reflect preference, helpfulness, style, or task conventions. Exact agreement and κ quantify correspondence with that target. They do not establish factual correctness, safety, or utility in every deployment. Similarly, repeated-evaluation consistency measures repeatability under a fixed condition, not an abstract stochastic property across arbitrary parameter settings.

6.3.2 Internal validity

The source-text remediation strengthens internal validity by binding controlled records to exact source text and preserving a documented source-precedence policy. The study avoids imputation, model substitution, and parser relaxation. Nevertheless, terminally unavailable observations remain. Fourteen terminal Claude observations were not treated as missing completely at random or missing at random. Complete-case estimators preserve transparency but may differ from an unavailable full outcome population.

6.3.3 External validity

The population consists of 80 MT-Bench questions represented as two turns, exact candidate answers, four judge families, and frozen protocols. Results should not automatically transfer to other tasks, languages, model releases, prompt templates, scoring rubrics, or provider routing configurations. The narrow RQ4/RQ5 treatments are especially unsuitable for generalized claims about all lengthy or formatted responses.

6.3.4 Statistical conclusion validity

The report gives denominators, coverage, and bootstrap intervals to prevent over-reading a point estimate. The DUAL_SWAP interval crossing zero is important. The RQ6 aggregate interval spanning 50% is important. Multi-Judge's positive interval is important, but it is conditional on a strict retained population and its own comparator. Multiple related descriptive breakdowns are presented as descriptive rather than as a series of independent confirmatory discoveries.

6.3.5 Operational validity

Provider failures, invalid responses, and persistence failures can distort an experiment if they are silently retried, relabeled, or dropped. The project records durable states and distinguishes physical completion, valid outcomes, and analytical eligibility. Live manual evaluation has a separate server-side safety gate and a separate telemetry class. A successful live manual trial cannot modify the frozen controlled evidence.

6.4 Practical recommendations

First, do not report an LLM evaluator only through one agreement percentage. Pair agreement with repeatability, order sensitivity, missingness, and coverage. Second, retain exact answer provenance; source labels are not a substitute for source identity. Third, expose the denominator and the rule that created it. Fourth, prefer protocols whose final API can fail closed rather than return a convenient hard-coded fallback. Fifth, distinguish a mitigation’s comparator and retained population before comparing its effect to another method. Finally, preserve evidence packages and release snapshots so that a final dashboard value can be independently recomputed later.

6.5 Future work

Future work could use a larger and more diverse prompt population, independently refreshed human preference labels, preregistered primary/secondary analyses, alternative consensus rules, and cost-aware operational comparisons. It could also study the source of the RQ3 pattern through controlled judge prompts and model versions. Such work should preserve the present project’s discipline: exact source text, explicit mapping of swapped presentations, visible terminal states, and modest language when causal explanation is not identified.

Chapter 7

System Implementation and Reproducibility

7.1 Application responsibilities

The project is implemented as a local research application, not merely a collection of analysis notebooks. The frontend offers a Leaderboard, Synthesis, Controlled Experiments, Bias Diagnostics, Qualitative Explorer, and Live Evaluation. The navigation intentionally makes the historical Leaderboard the default landing page while keeping final controlled results clearly accessible. The Synthesis page gives an executive account of all research questions. The Controlled Experiments page gives the detailed RQ evidence. Historical exploratory pages use their own data sources and are not presented as final RQ1–RQ7 evidence.

The backend separates API concerns from controlled evaluation, statistical analysis, specialized experiments, multi-judge procedures, release tooling, and historical provenance work. This package structure reduces the risk that a historical repair script becomes an accidental runtime dependency. It also makes reproducibility scripts identifiable as standalone tools rather than dead code merely because they are not imported by an API route.

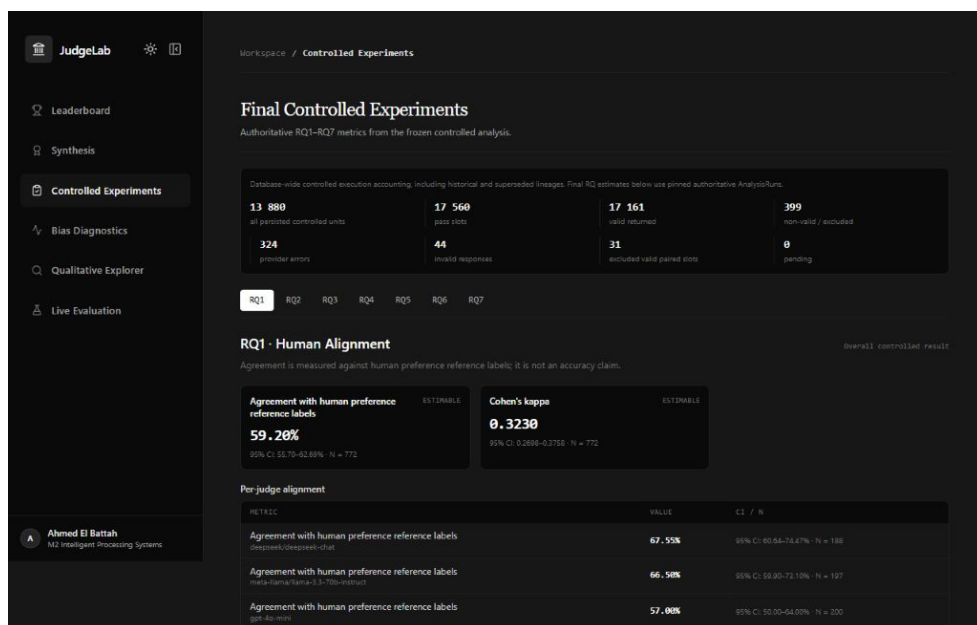


Figure 7.1: Local Controlled Experiments interface captured without any provider request. It displays analysis-run-pinned final evidence and distinguishes coverage from excluded slots.

7.2 API safety and analysis-run selection

The controlled-results API is deliberately DB-authoritative. It selects a uniquely matching analysis run by expected identity and required provenance, validates required metrics and confidence intervals, and fails closed if the selection is missing, duplicate, wrong-role, or malformed. RQ7 exposes DUAL_SWAP as PRIMARY and Multi-Judge Consensus as SECONDARY. It does not emit a synthetic direct DUAL_SWAP-versus-Multi-Judge delta. Focused tests cover missing run, duplicate run, wrong RQ, wrong role, missing metrics, missing confidence intervals, and missing provenance cases.

This design is a reproducibility feature as much as a software-engineering feature. A hard-coded front-end number would be easy to display but hard to audit. A fail-closed response makes an evidence break visible. The frontend uses API-derived values and presents the two mitigation families as complementary, each with its own comparator boundary.

7.3 Live Evaluation isolation

Live Evaluation is a local manual sandbox. Its provider execution gate is server-side and controlled through `ENABLE_LIVE_SANDBOX_PROVIDER_CALLS`; the safe example configuration defaults to false. When disabled, execution is rejected before any provider transport. When enabled in a local environment with credentials present, standard, calibrated, and ensemble modes can be used for an ad-hoc manual trial. The UI avoids population-level claims: it reports the current trial, a dual-pass result, or an order-sensitive change observed in this trial. It does not call a single trial a model-level bias finding or empirical debiasing result.

Most importantly, live results are classified separately from historical exploratory telemetry and frozen controlled evidence. They cannot modify final analysis runs or feed the final API. This is tested as a telemetry-isolation invariant.

7.4 Reproducibility packages

The repository preserves a historical Phase 11 evidence package and newer source-corrected and release-v4 packages. Historical artifacts are not deleted or silently reinterpreted; the report states which final source-corrected artifact is authoritative. Frozen verifier scripts check checksums, inventory, provenance, and result consistency. Release snapshots provide database restoration and API reconciliation checks. These procedures can be run provider-free because they validate persisted evidence rather than generating new judge outcomes.

Table 7.1: Reproducibility layers and their roles.

Layer	Role
Canonical data	Exact source text, question identity, answer identity, and source hashes.
Controlled database	Units, passes, attempts, terminal states, and analysis-run records.
Analysis artifact	Frozen estimator output, seeds, bootstrap policy, lineages, and numerical results.
Evidence packages	Checksums, inventories, and immutable copies for audit/reconstruction.
Release snapshots	Fresh database restore and API reconciliation without provider calls.
Automated tests	Route fail-closed behaviour, provenance, controlled metrics, and telemetry isolation.

7.5 Provider-free verification workflow

A fresh verification workflow begins by checking the canonical data hash and the source-corrected artifact identity. It then verifies the release package, restores the disposable release database, starts the backend against that restored database, and calls the controlled-results endpoint. The returned RQ1–RQ6 values and the two RQ7 components are reconciled with their authoritative analysis runs. Finally, the frontend tests, linting, type checks, and production build ensure the interface remains an accurate consumer of the API.

This workflow is intentionally capable of running without provider credit. It verifies the scientific evidence already collected, not the availability or present-day behavior of a provider. A later researcher can therefore distinguish a reproducibility failure from an execution-cost constraint.

7.6 Repository organization

The final repository keeps top-level backend essentials and semantic packages for API, core, data, evaluation, analysis, experiments, multi-judge work, release tooling, and historical

lineage. A small set of compatibility shims is retained only where immutable frozen verification paths demonstrably require a legacy import path. Standalone historical tools live outside active runtime areas. This organization makes the public workflow easier to identify while preserving the scripts needed to explain or validate historical and corrected releases.

7.7 Ethical and operational considerations

Automated judging affects which systems are selected, tuned, or trusted. Overstating an evaluator’s reliability can therefore propagate unfair or low-quality choices. The practical safeguards in this project are provenance, fail-closed API selection, visible coverage, no implicit imputation, and separation of exploratory/manual interaction from final evidence. These safeguards do not make an evaluator neutral or universally reliable. They make the limits of the reported result inspectable.

Chapter 8

Conclusion

This Master’s/PFE project developed and studied a reproducible LLM-as-a-Judge reliability laboratory. Its principal result is not that language-model judges are simply reliable or unreliable. Instead, the evidence identifies a profile. On the source-corrected final population, the judges reached 59.20% agreement with human preference reference labels and showed 96.53% fixed-condition repeated consistency. At the same time, 17.49% of decisive paired comparisons changed their mapped outcome after answer-order swapping. These results demonstrate why agreement alone is insufficient for evaluating an automatic judge.

The controlled redundant-length and presentation-format treatments did not obtain stable variant wins in their eligible populations. The source-family experiment did not support a uniform aggregate same-family preference, while revealing strong judge-level heterogeneity. RQ7 showed that mitigation must be presented with both agreement and coverage. DUAL_SWAP had a small, uncertain agreement change and a material coverage reduction. Strict Multi-Judge Consensus provided a positive secondary improvement relative to its own equal-weight individual-judge comparator on its retained pairs. Their effects are not direct head-to-head measurements.

The project also demonstrates an engineering lesson: final scientific results require an auditable path from exact source text to stored observations, analysis-run identities, release artifacts, API responses, and user interface. The source-text correction, durable terminal-state accounting, API fail-closed rules, and provider-free release validation make the numbers more than dashboard labels. They make the evidence inspectable.

The final recommendation is therefore practical and cautious. Use LLM judges as controlled instruments, not as unquestioned arbiters. Report human-reference agreement together

with repeatability, order sensitivity, treatment scope, coverage, and failure states. Keep mitigation comparators explicit. Preserve exact provenance and frozen artifacts. Under these conditions, automated judging can support scalable evaluation while retaining the methodological limits needed for credible research conclusions.

Appendix A

Frozen Protocol and Evidence Identity

A.1 Analysis policy

The final source-corrected analysis is governed by the following fixed policy: source precedence is SOURCE_CORRECT_REUSABLE, CORRECTED_ORIGINAL, RECOVERY_REPLACEMENT, then TERMINAL_MISSING; no imputation; no model substitution; no parser relaxation; 10,000 bootstrap iterations; seed 20260818 for controlled and RQ6 analyses; and seed 20260823 for RQ7 secondary Multi-Judge analysis. The scope is full-population source precedence. The analysis is a review of persisted data, with zero additional provider calls and zero additional spend.

A.2 Authoritative analysis-run identities

Table A.1: Authoritative source-corrected analysis-run identifiers.

Research ques- tion	AnalysisRun UUID
RQ1	115de122-6298-4aab-a255-dd5aedf9b3cc
RQ2	32f93918-f127-4cde-a0cd-6a73659daca7
RQ3	63995b48-a3a6-4386-b9b2-5f8bea3adfdb

Research ques- tion	AnalysisRun UUID
RQ4	d8eac555-6c46-49e7-ac19-8c271e719849
RQ5	067b3b9a-c739-41d9-8e2a-6d44478e6299
RQ6	416edb12-b0ba-4851-9aad-6e7ad4a01529
RQ7 primary DUAL_SWAP	c7057075-722a-4559-ac9b-d24e068d3769
RQ7 secondary Multi-Judge	feef94e7-ca66-4fdf-8cf1-a46edd6094b4

A.3 Manifest and release identities

The corrected manifest SHA-256 is 331d4498d5fe1b86d544366a3579e8f35344e518894350c0b8d339c551

The recovery manifest SHA-256 is 2fd7b64246cfba8421dc3d714fc272cef65214ecef1286c96318d0e65a

The counterbalanced RQ6 manifest SHA-256 is 27ec2c49ee8129802bd924e96734b4efb03249e506e7fbd9

The historical Multi-Judge execution manifest is retained separately, SHA-256 6c2fa4a926e458c598c7360

The historical Phase 11 evidence digest remains f1a1d6ffcb6f6fd5a5dd48f7b51a731d6b765a68ff76501

The corrected evidence digest is ce6354ab2c101cf505123754c17f99ad3aa5068d161bd2f9f863a9252538

The release-v4 digest is f2e72a6a0f4fee1aaf7eacf453c217957673a7229673bce96310827e9b681482.

These identifiers are reported for verification, not as substitute scientific metrics.

A.4 Claim boundary checklist

1. Report “human preference reference labels,” not ground truth.
2. Report the RQ3 mapped decisive flip rate, not a raw slot-preference statistic.
3. Describe RQ4 as a controlled redundant-text effect and RQ5 as a controlled presentation-format effect.
4. Report RQ6 as a counterbalanced source-family association with heterogeneous judge-level patterns, not proven self-bias.

-
5. State DUAL_SWAP and Multi-Judge comparators separately and do not claim direct superiority.
 6. Treat live manual trials as sandbox observations, never as additions to frozen controlled evidence.

Appendix B

Numerical Result Ledger

Table B.1: Compact source-corrected final result ledger.

RQ	Metric	Result	Population	Interpretation boundary
RQ1	Exact agreement	$457/772 = 59.20\%$; CI $55.70\text{--}62.69\%$; $\kappa =$.3230	772 valid analytical observations	Agreement with human preference refer- ence labels only.
RQ2	Strict repeated consistency	$7,119/7,375 = 96.53\%$	1,475 strict valid cells	Fixed-condition repetition; not a general temperature effect.
RQ3	Decisive mapped flip rate	$96/549 = 17.49\%$	549 decisive AB/BA mapped pairs	Answer-order sensitivity under this pro- tocol.
RQ3	All paired disagreement	$193/759 = 25.43\%$	759 valid paired out- comes	Descriptive broader disagreement, in- cludes non-decisive differences.
RQ4	Stable redundant-variant wins	$0/682 = 0.00\%$	682 eligible pairs	Narrow frozen redundant-text treatment only.
RQ5	Stable format-variant wins	$0/716 = 0.00\%$	716 eligible pairs	Narrow frozen list-prefix/presentation treatment only.
RQ6	Stable same-family prefer- ence	$81/159 = 50.94\%$; CI $42.77\text{--}58.49\%$	159 stable decisive; 33.13% coverage	Counterbalanced association; heteroge- neous judges; no causal claim.
RQ7 primary	Baseline agreement	$405/597 = 67.84\%$	597 retained matched units	Primary within-judge comparator.
RQ7 primary	DUAL_SWAP agreement	$409/597 = 68.51\%$; delta $+0.67$ pp; CI -0.17 to $+1.68$	597 retained matched units	Small uncertain gain; not a universal mit- igation claim.
RQ7 primary	Coverage	96.87% to 75.84% ; delta -21.03 pp	799 linked triplets	Coverage is part of result.

RQ	Metric	Result	Population	Interpretation boundary
RQ7 secondary	Strict consensus agreement	808/1,125 = 71.82%	1,125/1,611 = 69.83% retained	Secondary cross-judge aggregation.
RQ7 secondary	Matched delta	+5.02 pp; CI +4.16 to +5.89	Same retained pairs vs equal-weight in- dividual baseline (66.80%)	Not directly comparable with DUAL_SWAP.

B.1 Coverage and exclusions

The ledger intentionally gives different denominators for different RQs. A research question that requires a decisive AB/BA comparison cannot use the denominator from a repetition experiment; a strict consensus result cannot use a DUAL_SWAP retained denominator. Using a common denominator simply for visual uniformity would be scientifically incorrect. Complete-case analysis prioritizes a transparent eligibility rule over an artificial 100% completion claim.

B.2 Historical comparison boundary

Historical analysis runs remain available for audit and comparison, but they are not the authoritative values in this report. The source-corrected final lineage retained the direction of the historical RQ effects while correcting the underlying source text and final denominators. The report therefore cites only the source-corrected analysis-run identities in Table A.1.

Appendix C

Provider-Free Reproducibility Guide

C.1 Purpose

This appendix describes how a reviewer can validate the final evidence without spending provider credit. It is not an instruction to rerun paid experiments. The review begins from the canonical source-corrected data and frozen release/evidence packages already contained in the repository.

C.2 Recommended sequence

1. Confirm the repository revision and inspect the working tree for unexpected changes.
2. Verify the canonical source-corrected data hash and the 1,568-record/zero-unresolved accounting.
3. Verify the source-corrected full-population analysis artifact identity and the result ledger.
4. Run the corrected-evidence and release-v4 verification tools.
5. Restore the release-v4 database into a disposable PostgreSQL instance.
6. Start the backend against that disposable database only and call the controlled-results endpoint.
7. Reconcile returned RQ1–RQ6 values and the RQ7 primary/secondary sections with the stored authoritative analysis runs.

8. Run backend tests, frontend tests, linting, TypeScript checks, and the production build.

C.3 What a successful check establishes

Successful verification establishes that the checked artifact and database snapshot have the expected bytes, that the backend can select the intended analysis records, and that the user interface can consume the API. It does not establish that a provider would behave identically today, because the workflow purposefully makes no provider request. This distinction is central: scientific reproducibility of stored evidence is different from repeat execution of a live model service.

C.4 Failure interpretation

If a checksum fails, the appropriate response is to stop and determine whether the artifact changed. If an analysis run is missing or duplicate, the controlled-results API should fail closed. If a historical explorer has missing answer provenance, it should return unavailable provenance rather than fabricate text. If a live sandbox request is disabled, it should be rejected before provider transport. These failures are protective signals, not reasons to replace stored values with constants.

C.5 Preservation recommendations

For archival use, preserve the repository revision, canonical source-corrected data, all evidence packages, release dumps, manifest hashes, and this report PDF. Preserve the distinction between historical Phase 11 evidence and newer source-corrected final evidence. When new research is performed, create a new named lineage; do not overwrite the identities summarized here.

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